

Primitive Integer Solutions of

$$3Y^3 + Z^3 = X^4$$

July 17, 2026

Abstract

We determine every primitive integral solution of

$$3y^3 + z^3 = x^4.$$

The answer consists of four ordered triples:

$$(\pm 1, 0, 1), \quad (\pm 895, 6202, -4199).$$

The proof factors $z + y\sqrt[3]{3}$ in the class-number-one cubic field $\mathbb{Q}(\sqrt[3]{3})$, reduces the unit ambiguity to two square classes, and parametrizes the two resulting quadratic coefficient conditions. The final stage is an exact, computer-assisted determination of the rational points on three explicit genus-two curves. All algebraic identities, local congruences, finite-field point counts, and numerical verifications are recorded explicitly. The genus-two calculation is isolated as a single computational proposition, so that the logical dependence of the proof is completely transparent.

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1 Statement of the result

Definition 1.1. *An integral triple (x, y, z) is called primitive if*

$$\gcd(x, y, z) = 1.$$

The gcd is understood to be nonnegative.

Theorem 1.2 (Complete classification). *The primitive integer solutions of*

$$3y^3 + z^3 = x^4 \tag{1}$$

are exactly

$$(x, y, z) = (\pm 1, 0, 1)$$

and

$$(x, y, z) = (\pm 895, 6202, -4199).$$

The large solution is genuine:

$$3 \cdot 6202^3 - 4199^3 = 641641050625 = 895^4,$$

and

$$\gcd(895, 6202, 4199) = 1.$$

The remainder of the paper proves that no other primitive triple exists.

Remark 1.3 (Nature of the proof). *The proof is unconditional but computer-assisted. The only machine-assisted input is proposition 5.1, which gives complete rational-point sets on three specified genus-two curves by exact two-descent, Jacobian arithmetic, quadratic Chabauty, and the Mordell–Weil sieve. No bounded search is used to infer completeness. Every other step is proved directly below.*

2 Elementary restrictions imposed by primitivity

Lemma 2.1 (Pairwise coprimality). *Let (x, y, z) be a primitive solution of (1). Then*

$$\gcd(x, y) = \gcd(x, z) = \gcd(y, z) = 1.$$

Proof. Let p be a prime.

If $p \mid x$ and $p \mid y$, then (1) gives $p \mid z$. If $p \mid y$ and $p \mid z$, then it gives $p \mid x$.

Suppose that $p \mid x$ and $p \mid z$. If $p \neq 3$, then $p \mid 3y^3$, hence $p \mid y$. If $p = 3$, then $3 \mid x, z$, so x^4 is divisible by 3^4 and z^3 is divisible by 3^3 . Therefore

$$3y^3 = x^4 - z^3$$

is divisible by 27, whence $9 \mid y^3$ and therefore $3 \mid y$. Thus in every case, a prime dividing any two coordinates divides all three, contrary to primitivity. This proves all three pairwise gcd assertions. $\square$

Corollary 2.2 (The prime 3). *For every primitive solution of (1),*

$$3 \nmid xz \quad \text{and} \quad z \equiv 1 \pmod{3}.$$

In particular, $x \neq 0$.

Proof. If $3 \mid x$, then reduction of (1) modulo 3 gives $3 \mid z$, contradicting lemma 2.1. Thus $3 \nmid x$, and the same congruence gives $3 \nmid z$. Since $x^4 \equiv 1 \pmod{3}$ and $z^3 \equiv z \pmod{3}$, equation (1) yields $z \equiv 1 \pmod{3}$. Finally, if $x = 0$, then $z^3 = -3y^3$. Comparing 3-adic valuations gives $3v_3(z) = 1 + 3v_3(y)$, an impossibility. Hence $x \neq 0$. $\square$

3 Arithmetic of the pure cubic field

Put

$$\alpha = \sqrt[3]{3}, \quad K = \mathbb{Q}(\alpha).$$

The three embeddings of K into $\mathbb{C}$ send α to $\alpha, \omega\alpha, \omega^2\alpha$, where $\omega = e^{2\pi i/3}$.

3.1 The ring of integers and the norm

Lemma 3.1 (Integral basis). *The ring of integers of K is*

$$\mathcal{O}_K = \mathbb{Z}[\alpha],$$

and the field discriminant is

$$\Delta_K = -243 = -3^5.$$

Proof. The discriminant of the power basis $1, \alpha, \alpha^2$ equals the discriminant of $T^3 - 3$, namely $-27 \cdot 3^2 = -243$. Consequently, a prime dividing the index $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$ must be 3.

It remains to prove maximality at 3. The polynomial $T^3 - 3$ is Eisenstein at 3, so $K \otimes_{\mathbb{Q}} \mathbb{Q}_3$ is a totally ramified cubic extension of $\mathbb{Q}_3$. Let w be its valuation normalized by

$$w(\alpha) = 1, \quad w(3) = 3.$$

Every element of $K \otimes \mathbb{Q}_3$ has a unique expression

$$u = c_0 + c_1\alpha + c_2\alpha^2, \quad c_i \in \mathbb{Q}_3.$$

The valuations of the three nonzero summands are

$$3v_3(c_0), \quad 3v_3(c_1) + 1, \quad 3v_3(c_2) + 2.$$

They are pairwise distinct modulo 3, so the terms of least valuation cannot cancel. If u is integral, then $w(u) \geq 0$, and hence each of the three displayed valuations is nonnegative. Since each $v_3(c_i)$ is an integer, this implies $c_i \in \mathbb{Z}_3$ for all i . Therefore the local integral closure is $\mathbb{Z}_3[\alpha]$, so the index is not divisible by 3. The index is therefore 1, proving the assertion. $\square$

Lemma 3.2 (Norm formula). *For $a, b, c \in \mathbb{Q}$,*

$$N_{K/\mathbb{Q}}(a + b\alpha + c\alpha^2) = a^3 + 3b^3 + 9c^3 - 9abc.$$

Proof. Multiplication by $a + b\alpha + c\alpha^2$ on the basis $1, \alpha, \alpha^2$ is represented by

$$\begin{pmatrix} a & 3c & 3b \\ b & a & 3c \\ c & b & a \end{pmatrix}.$$

Its determinant is $a^3 + 3b^3 + 9c^3 - 9abc$, and this determinant is the field norm. $\square$

3.2 Class number one

We include the precise Minkowski estimate needed below.

Lemma 3.3 (Minkowski class bound in signature $(1, 1)$). *Let L be a cubic number field with one real embedding, one pair of complex embeddings, and discriminant Δ_L . Every ideal class of L contains an integral ideal $\mathfrak{b}$ satisfying*

$$N(\mathfrak{b}) \leq \frac{8}{9\pi} \sqrt{|\Delta_L|}.$$

Proof. Embed a nonzero integral ideal $\mathfrak{a}$ as a lattice in $\mathbb{R} \times \mathbb{C}$. Its covolume is

$$2^{-1} \sqrt{|\Delta_L|} N(\mathfrak{a}).$$

For $T > 0$, consider the centrally symmetric convex body

$$B_T = \{(u, z) \in \mathbb{R} \times \mathbb{C} : |u| + 2|z| \leq T\}.$$

A direct integration gives

$$\text{vol}(B_T) = 2 \int_0^T \pi \left(\frac{T-u}{2} \right)^2 du = \frac{\pi T^3}{6}.$$

Minkowski's convex-body theorem therefore gives a nonzero $\xi \in \mathfrak{a}$ as soon as

$$\frac{\pi T^3}{6} > 2^3 \cdot 2^{-1} \sqrt{|\Delta_L|} N(\mathfrak{a}),$$

that is, as soon as

$$T^3 > \frac{24}{\pi} \sqrt{|\Delta_L|} N(\mathfrak{a}).$$

For such a point, the arithmetic–geometric mean inequality applied to $|u|, |z|, |z|$ gives

$$|N_{L/\mathbb{Q}}(\xi)| = |u| |z|^2 \leq \left(\frac{|u| + 2|z|}{3} \right)^3 \leq \frac{T^3}{27}.$$

Letting T decrease to the threshold gives

$$|N(\xi)| \leq \frac{8}{9\pi} \sqrt{|\Delta_L|} N(\mathfrak{a}).$$

Because $(\xi) \subseteq \mathfrak{a}$, the ideal $\mathfrak{b} = (\xi)\mathfrak{a}^{-1}$ is integral, lies in the inverse class of $\mathfrak{a}$, and satisfies

$$N(\mathfrak{b}) = \frac{|N(\xi)|}{N(\mathfrak{a})} \leq \frac{8}{9\pi} \sqrt{|\Delta_L|}.$$

As inversion permutes ideal classes, the result follows. $\square$

Proposition 3.4 (Class number). *The field $K = \mathbb{Q}(\sqrt[3]{3})$ has class number 1.*

Proof. By lemmas 3.1 and 3.3, every ideal class contains an integral ideal of norm at most

$$\frac{8}{9\pi}\sqrt{243} = \frac{8\sqrt{3}}{\pi} < 4.42.$$

Thus it suffices to consider ideal norms 2, 3, 4.

The following elements have the indicated positive absolute norms:

$$N(\alpha - 1) = 2, \quad N(\alpha) = 3, \quad N(\alpha^2 + \alpha + 1) = 4.$$

Modulo 2,

$$T^3 - 3 \equiv T^3 + 1 = (T + 1)(T^2 + T + 1),$$

so the prime ideals above 2 have norms 2 and 4; they are generated by $\alpha - 1$ and $\alpha^2 + \alpha + 1$, respectively. An ideal of norm 4 is either the degree-two prime above 2 or the square of the norm-2 prime, and is therefore principal. The unique prime above 3 is (α) and is principal. Consequently every ideal of norm at most 4 is principal, so every ideal class is trivial. $\square$

3.3 The unit group

Define

$$\eta = \alpha^2 - 2, \quad \varepsilon = 4 + 3\alpha + 2\alpha^2.$$

A direct multiplication gives

$$(\alpha^2 - 2)(4 + 3\alpha + 2\alpha^2) = 1,$$

so $\varepsilon = \eta^{-1}$. Also lemma 3.2 gives $N(\eta) = N(\varepsilon) = 1$.

Lemma 3.5 (All units). *The unit group of K is*

$$\mathcal{O}_K^\times = \{\pm \varepsilon^n : n \in \mathbb{Z}\}.$$

Its norm-1 subgroup is

$$\{\eta^n : n \in \mathbb{Z}\}.$$

Proof. Let $u = a + b\alpha + c\alpha^2$ be a unit, and let σ_1 be the real embedding. Since $|N(u)| = 1$, the two nonreal conjugates have common absolute value $|\sigma_1(u)|^{-1/2}$.

First consider a unit satisfying

$$1 \leq |\sigma_1(u)| < \varepsilon.$$

Fourier inversion among the three embeddings gives

$$\begin{aligned} 3a &= \sigma_1(u) + \sigma_2(u) + \sigma_3(u), \\ 3\alpha b &= \sigma_1(u) + \omega^2\sigma_2(u) + \omega\sigma_3(u), \\ 3\alpha^2 c &= \sigma_1(u) + \omega\sigma_2(u) + \omega^2\sigma_3(u). \end{aligned}$$

Because the two complex conjugates have absolute value at most 1, and because

$$\varepsilon = 4 + 3\alpha + 2\alpha^2 < 13,$$

we obtain the sharper numerical bounds

$$|a| < 5, \quad |b| < 4, \quad |c| < 3.$$

Thus

$$|a| \leq 4, \quad |b| \leq 3, \quad |c| \leq 2.$$

Substitution into the exact norm equation

$$a^3 + 3b^3 + 9c^3 - 9abc = \pm 1$$

leaves precisely the following six triples:

(a, b, c)	$a + b\alpha + c\alpha^2$	N
$(-4, -3, -2)$	$-\varepsilon$	-1
$(-2, 0, 1)$	η	1
$(-1, 0, 0)$	-1	-1
$(1, 0, 0)$	1	1
$(2, 0, -1)$	$-\eta$	-1
$(4, 3, 2)$	ε	1

The interval $1 \leq |\sigma_1(u)| < \varepsilon$ contains only $u = \pm 1$ from this list: $|\eta| = \varepsilon^{-1} < 1$, while $|\varepsilon|$ is the excluded upper endpoint.

Now let u be any unit. Choose $n \in \mathbb{Z}$ such that

$$1 \leq |\sigma_1(u\varepsilon^{-n})| < \varepsilon.$$

The preceding paragraph gives $u\varepsilon^{-n} = \pm 1$, hence $u = \pm \varepsilon^n$. Finally, $N(-1) = -1$ because $[K : \mathbb{Q}] = 3$, while $N(\varepsilon) = 1$. Therefore the norm-1 units are exactly the powers of ε , equivalently the powers of $\eta = \varepsilon^{-1}$. $\square$

4 Fourth-power ideal descent

Let (x, y, z) be a primitive solution and put

$$\beta = z + y\alpha \in \mathcal{O}_K.$$

By lemma 3.2,

$$N(\beta) = z^3 + 3y^3 = x^4.$$

Lemma 4.1 (Prime-ideal support of β). *Let p be a rational prime dividing x . Then $p \neq 3$, and exactly one prime ideal $\mathfrak{p}$ above p divides (β) . This prime has residue degree 1, and*

$$v_{\mathfrak{p}}(\beta) = 4v_p(x).$$

No prime above 3 divides (β) .

Proof. By lemma 2.1 and corollary 2.2, $p \neq 3$ and $p \nmid yz$. Reducing the norm equation modulo p gives

$$(-zy^{-1})^3 \equiv 3 \pmod{p}.$$

Set $t \equiv -zy^{-1} \pmod{p}$. Since $p \neq 3$ and $t \neq 0$, the root t of $T^3 - 3$ modulo p is simple. Hence

$$\mathfrak{p} = (p, \alpha - t)$$

is a degree-one prime above p .

If any prime $\mathfrak{q} \mid p$ divides β , then in its residue field we have $z + y\alpha = 0$, so $\alpha = t \in \mathbb{F}_p$. Therefore $\mathfrak{q} = \mathfrak{p}$. Thus $\mathfrak{p}$ is the unique prime above p occurring in (β) . Taking p -adic valuations of the ideal norm now gives

$$4v_p(x) = v_p(N(\beta)) = f(\mathfrak{p}/p)v_{\mathfrak{p}}(\beta) = v_{\mathfrak{p}}(\beta).$$

Finally, the unique prime above 3 is (α) . Modulo this prime, $\beta \equiv z$, and $3 \nmid z$ by corollary 2.2; hence it does not divide (β) . $\square$

Proposition 4.2 (Reduction to two unit square classes). *There exist an element*

$$\delta = a + b\alpha + c\alpha^2 \in \mathcal{O}_K$$

and an exponent $e \in \{0, 1\}$ such that

$$z + y\alpha = \eta^e \delta^2, \tag{2}$$

with

$$N(\delta) = x^2 \quad \text{and} \quad \gcd(a, b, c) = 1.$$

Proof. By lemma 4.1, every prime-ideal exponent in (β) is a multiple of 4. Hence

$$(\beta) = \mathfrak{a}^4$$

for an integral ideal $\mathfrak{a}$. By proposition 3.4, $\mathfrak{a} = (\gamma)$, and therefore

$$\beta = u\gamma^4$$

for a unit u . Taking norms and using $|N(\gamma)| = N(\mathfrak{a}) = |x|$ gives $N(u) = 1$. By lemma 3.5, $u = \eta^n$ for some $n \in \mathbb{Z}$.

Write $n = e + 2k$ with $e \in \{0, 1\}$, and put

$$\delta = \eta^k \gamma^2.$$

Then (2) holds, and

$$N(\delta) = N(\gamma)^2 = x^2.$$

It remains to prove coefficient primitivity. If an integer $d > 1$ divided a, b, c , then $\delta = d\delta_0$ for some $\delta_0 \in \mathcal{O}_K$, so (2) would show that every coefficient of $z + y\alpha$ in the integral basis $1, \alpha, \alpha^2$ is divisible by d^2 . Thus $d^2 \mid y$ and $d^2 \mid z$, contradicting $\gcd(y, z) = 1$ from lemma 2.1. $\square$

For later use, write

$$\delta^2 = A + B\alpha + C\alpha^2.$$

A direct expansion gives

$$A = a^2 + 6bc, \quad B = 2ab + 3c^2, \quad C = b^2 + 2ac. \tag{3}$$

5 The three genus-two computations

We use weighted projective coordinates $[R : S : W] \in \mathbb{P}(1, 1, 3)$, with

$$[R : S : W] = [\lambda R : \lambda S : \lambda^3 W] \quad (\lambda \in \mathbb{Q}^\times).$$

Define the curves

$$C_A : \quad W^2 = R^6 + 60R^3S^3 - 72S^6, \quad (4)$$

$$C_B : \quad W^2 = -8R^6 - 60R^3S^3 + 9S^6, \quad (5)$$

and define

$$\begin{aligned} F(M, N) = & 720M^6 - 1944M^5N + 1620M^4N^2 + 60M^3N^3 \\ & - 810M^2N^4 + 432MN^5 - 73N^6, \end{aligned} \quad (6)$$

with

$$C_F : \quad W^2 = -F(M, N). \quad (7)$$

Proposition 5.1 (Exact rational-point classification). *The complete rational-point sets are*

$$C_A(\mathbb{Q}) = \{[1 : 0 : 1], [1 : 0 : -1]\},$$

$$C_B(\mathbb{Q}) = \{[0 : 1 : 3], [0 : 1 : -3]\},$$

and

$$C_F(\mathbb{Q}) = \{[1 : 2 : \pm 4], [2 : 3 : \pm 3], [2 : 7 : \pm 895]\}.$$

Exact computational certificate. This is the sole computer-assisted step. We record the defining data and the exact arithmetic certificate.

(i) Good reduction and finite-field checks. The discriminant of each dehomogenized sextic is

$$2^{18}3^{25}.$$

Thus 7 and 11 are primes of good reduction. Direct finite-field enumeration gives the following table. The Jacobian orders are obtained from

$$\#J(\mathbb{F}_p) = \frac{\#C(\mathbb{F}_p)^2 + \#C(\mathbb{F}_{p^2})}{2} - p.$$

C	p	$\#C(\mathbb{F}_p)$	$\#C(\mathbb{F}_{p^2})$	$\#J(\mathbb{F}_p)$
C_A	7	8	64	57
C_A	11	12	94	108
C_B	7	8	64	57
C_B	11	12	94	108

An exact 2-descent gives

$$\dim_{\mathbb{F}_2} \text{Sel}^{(2)}(J_A/\mathbb{Q}) = \dim_{\mathbb{F}_2} \text{Sel}^{(2)}(J_B/\mathbb{Q}) = 0.$$

Hence both Jacobians have Mordell–Weil rank 0.

For C_A , in affine coordinates,

$$Y^2 = (X^3 + 30)^2 - 972.$$

Let I_+ and I_- be the two rational points at infinity, distinguished by $Y/X^3 \rightarrow +1$ and $Y/X^3 \rightarrow -1$. Then

$$\operatorname{div}(Y - (X^3 + 30)) = 3I_+ - 3I_-.$$

Thus $[I_+ - I_-]$ is a nonzero 3-torsion point. Since rational torsion injects under good reduction and

$$\gcd(57, 108) = 3,$$

we obtain

$$J_A(\mathbb{Q}) \cong \mathbb{Z}/3\mathbb{Z}.$$

Embedding C_A in J_A by $P \mapsto [P - I_-]$ gives the two displayed points and no third point. Indeed, a third point would give an effective divisor linearly equivalent to $2I_+$ but different from $2I_+$, forcing $\ell(2I_+) \geq 2$. This is impossible because I_+ is not a Weierstrass point on a genus-two hyperelliptic curve.

For C_B , put

$$P_+ = (0, 3), \quad P_- = (0, -3), \quad q(X) = 3 - 10X^3.$$

The identity

$$(-8X^6 - 60X^3 + 9) - q(X)^2 = -108X^6$$

gives

$$\operatorname{div}\left(\frac{Y - q(X)}{X^3}\right) = 3P_+ - 3P_-.$$

The same reduction argument therefore gives $J_B(\mathbb{Q}) \cong \mathbb{Z}/3\mathbb{Z}$, and the same non-Weierstrass divisor argument gives exactly P_+ and P_- .

(ii) The curve C_F . Make the unimodular substitution

$$M = -U - V, \quad N = -2U - V.$$

Then

$$F(-U - V, -2U - V) = -H(U, V),$$

where

$$\begin{aligned} H(U, V) = & 16U^6 - 24U^5V + 60U^4V^2 - 20U^3V^3 \\ & - 30U^2V^4 + 12UV^5 - 5V^6. \end{aligned}$$

Thus C_F is rationally isomorphic to $C_H : W^2 = H(U, V)$. Let $I_\pm$ be the two points at infinity of C_H , and set

$$D_0 = [I_- - I_+], \quad D_1 = [(1, 3) - I_+].$$

Exact two-descent followed by saturation gives

$$J_H(\mathbb{Q}) = \mathbb{Z}D_0 \oplus \mathbb{Z}D_1.$$

The torsion subgroup is trivial: the exact point counts are

$$\#J_H(\mathbb{F}_7) = 57, \quad \#J_H(\mathbb{F}_{11}) = 82,$$

and $\gcd(57, 82) = 1$.

The identity

$$\begin{aligned} H(T, 1) - (4T^3 - 3T^2 - 24T + 20)^2 \\ = 81(T - 1)^3(3T + 5) \end{aligned} \quad (8)$$

exhibits the six rational points

$$I_{\pm}, \quad (1, \pm 3), \quad \left(-\frac{5}{3}, \pm \frac{895}{27}\right).$$

A quadratic-Chabauty computation at $p = 7$, combined with the Mordell–Weil sieve for the saturated group generated by D_0, D_1 , is exhaustive. The eight residue disks of $C_H(\mathbb{F}_7)$ and their certified Strassmann bounds are

disk	I_+	I_-	$(1, 3)$	$(1, -3)$	$(3, 1)$	$(3, -1)$	$(0, 3)$	$(0, -3)$
# rational points	1	1	1	1	1	1	0	0.

The two points with $T = -5/3$ reduce to the disks with $T = 3$ modulo 7. Hence the six displayed points are all of $C_H(\mathbb{Q})$.

Finally, the inverse linear change is

$$U = M - N, \quad V = N - 2M.$$

The six points of $C_H(\mathbb{Q})$ transform into

$$[1 : 2 : \pm 4], \quad [2 : 3 : \pm 3], \quad [2 : 7 : \pm 895]$$

on C_F , proving the asserted list. $\square$

Remark 5.2 (What the certificate proves). *The descent and Chabauty calculations in proposition 5.1 are exact computations in number fields, finite fields, Jacobians, and 7-adic power series. The point lists are therefore not conclusions drawn from a height search. The finite-field orders, polynomial identities, and all displayed points can be checked independently by the elementary verification script supplied with this manuscript.*

6 The first unit class

Assume first that $e = 0$ in (2). Then

$$z + y\alpha = \delta^2.$$

By (3), the vanishing of the α^2 coefficient is

$$b^2 + 2ac = 0. \quad (9)$$

Lemma 6.1 (Primitive parametrization of the first conic). *Every primitive integral solution (a, b, c) of (9) has, up to an overall sign $\sigma \in \{\pm 1\}$, exactly one of the forms*

$$(a, b, c) = \sigma(r^2, 2rs, -2s^2), \quad (10)$$

where $\gcd(r, s) = 1$ and r is odd, or

$$(a, b, c) = \sigma(2r^2, 2rs, -s^2), \quad (11)$$

where $\gcd(r, s) = 1$ and s is odd.

Proof. Equation (9) shows that b is even; write $b = 2d$. Then

$$ac = -2d^2.$$

If a prime divided both a and c , the equation would force it to divide b , contrary to primitivity. Thus $\gcd(a, c) = 1$.

Every odd prime occurs in a or c to an even exponent. Exactly one of a, c is even unless one is zero, and the even one is twice a square up to sign because the 2-adic valuation of ac is odd. If a is odd, write $a = \sigma r^2$ and $c = -2\sigma s^2$; then $d = \pm rs$, and the sign of d is absorbed by changing the sign of r or s . This gives (10), with r odd. If c is odd, the same argument gives (11), with s odd. Coprimality of r, s is equivalent to primitivity. The degenerate cases $b = 0$ are included by $s = 0$ in (10) and by $r = 0$ in (11). $\square$

For (10), the norm is

$$N(\delta) = \sigma(r^6 + 60r^3s^3 - 72s^6). \quad (12)$$

For (11), it is

$$N(\delta) = \sigma(8r^6 + 60r^3s^3 - 9s^6). \quad (13)$$

These identities follow by substitution into lemma 3.2.

Lemma 6.2 (The norm signs in the first unit class). *In (12), the equality $N(\delta) = x^2$ forces $\sigma = 1$. In (13), it forces $\sigma = -1$.*

Proof. The square residues modulo 16 are 0, 1, 4, 9.

In (12), r is odd. If s is even, then

$$r^6 + 60r^3s^3 - 72s^6 \equiv 1 \text{ or } 9 \pmod{16},$$

so its negative is 15 or 7 modulo 16. If s is odd, the form is 5 or 13 modulo 16, so its negative is 11 or 3. Thus the negative of the form is never a square modulo 16, proving $\sigma = 1$.

In (13), s is odd. If r is even, the form is 7 or 15 modulo 16; if r is odd, it is 3 or 11 modulo 16. It is therefore never a square, so $\sigma = -1$. $\square$

We now have

$$x^2 = r^6 + 60r^3s^3 - 72s^6$$

in the first family, and

$$x^2 = -8r^6 - 60r^3s^3 + 9s^6$$

in the second. By proposition 5.1, the first equation has only $[r : s : x] = [1 : 0 : \pm 1]$, while the second has only $[r : s : x] = [0 : 1 : \pm 3]$.

For the first point, (10) gives $\delta = 1$ up to sign, so

$$z + y\alpha = 1.$$

Thus

$$(x, y, z) = (\pm 1, 0, 1).$$

For the second point, (11) with the required sign gives $\delta = \alpha^2$ up to sign. Hence

$$z + y\alpha = \alpha^4 = 3\alpha,$$

so

$$(x, y, z) = (\pm 3, 3, 0),$$

which is not primitive. Therefore the first unit class contributes exactly

$$(\pm 1, 0, 1).$$

7 The second unit class

Assume now that $e = 1$. Since $\eta = \alpha^2 - 2$, equations (3) and (2) give

$$\eta\delta^2 = (3B - 2A) + (3C - 2B)\alpha + (A - 2C)\alpha^2.$$

The α^2 coefficient must vanish, so

$$a^2 - 4ac - 2b^2 + 6bc = 0. \quad (14)$$

Lemma 7.1 (Primitive parametrization of the second conic). *Let (a, b, c) be a primitive integral solution of (14). Then there exist coprime integers m, n , an overall sign $\sigma \in \{\pm 1\}$, and*

$$g = \begin{cases} 1, & n \text{ odd}, \\ 2, & n \text{ even}, \end{cases}$$

such that

$$(a, b, c) = \frac{\sigma}{g}(2n(3m - 2n), 2m(3m - 2n), 2m^2 - n^2). \quad (15)$$

Conversely, every triple in (15) is a primitive integral solution of (14).

Proof. In projective coordinates, the conic contains $[0 : 0 : 1]$. For a point with $a \neq 0$, write $b/a = m/n$ in lowest terms. Substitution into (14) gives

$$\frac{c}{a} = \frac{2m^2 - n^2}{2n(3m - 2n)}.$$

Clearing denominators yields the displayed homogeneous parametrization. The point $[0 : 3 : 1]$ occurs when $n = 0$, and the base point $[0 : 0 : 1]$ occurs when $3m - 2n = 0$, so no projective point is omitted.

Let

$$a_0 = 2n(3m - 2n), \quad b_0 = 2m(3m - 2n), \quad c_0 = 2m^2 - n^2$$

with $\gcd(m, n) = 1$. No odd prime divides all three. Indeed, if an odd prime p divides a_0, b_0 , then $p \mid 3m - 2n$; because p cannot divide both m, n , the congruence $3m \equiv 2n \pmod{p}$ gives

$$4c_0 = 8m^2 - 4n^2 \equiv -m^2 \not\equiv 0 \pmod{p}.$$

If n is odd, then c_0 is odd, so the gcd is 1. If n is even, then m is odd; in that case b_0 and c_0 are divisible by 2 but not by 4, so the gcd is exactly 2. Division by g therefore gives a primitive triple. The converse follows by the projective parametrization and the uniqueness of a primitive integral representative up to sign. $\square$

Substitution of the unscaled triple into the norm formula gives precisely the sextic F from (6):

$$N(\delta) = \frac{\sigma F(m, n)}{g^3} = x^2. \quad (16)$$

Lemma 7.2 (The even- n case is impossible). *There is no solution of (16) with n even.*

Proof. If n is even, then $g = 2$ and m is odd. Equation (16) gives

$$(4x)^2 = 2\sigma F(m, n).$$

Write $n = 2k$. Modulo 32,

$$F(m, 2k) \equiv 16(1 - m^5k + m^4k^2).$$

Since m is odd, the factor in parentheses is odd, whether k is even or odd. Hence

$$F(m, n) \equiv 16 \pmod{32}$$

and therefore

$$2\sigma F(m, n) \equiv 32 \pmod{64}.$$

But a square divisible by 16 is congruent to 0 or 16 modulo 64, never 32. This contradiction proves the claim. $\square$

Henceforth n is odd and $g = 1$.

Lemma 7.3 (The positive norm sign is impossible). *For coprime m, n with n odd, $F(m, n)$ is never a square. Consequently (16) forces*

$$\sigma = -1 \quad \text{and} \quad x^2 = -F(m, n).$$

Proof. Modulo 27,

$$F(m, n) \equiv 18m^6 + 6m^3n^3 + 8n^6. \tag{17}$$

The square residues modulo 27 are

$$0, 1, 4, 7, 9, 10, 13, 16, 19, 22, 25.$$

If $3 \mid n$, then $3 \nmid m$ and (17) gives $F(m, n) \equiv 18 \pmod{27}$, not a square. If $3 \mid m$, then $3 \nmid n$ and $F(m, n)$ is congruent to one of 8, 17, 26, again not a square.

Finally suppose $3 \nmid mn$. Divide (17) by the square n^6 and put

$$t \equiv (mn^{-1})^3 \pmod{27}.$$

The unit cubes modulo 27 are

$$1, 8, 10, 17, 19, 26.$$

For these six values,

$$18t^2 + 6t + 8 \equiv 5 \text{ or } 20 \pmod{27},$$

neither of which is a square. Thus $F(m, n)$ is never a square. Since $x^2 = \sigma F(m, n)$, the sign must be $\sigma = -1$. $\square$

By proposition 5.1, the primitive projective solutions of

$$x^2 = -F(m, n)$$

are

$$[m : n : x] = [1 : 2 : \pm 4], \quad [2 : 3 : \pm 3], \quad [2 : 7 : \pm 895].$$

The first has n even and was excluded by lemma 7.2.

For $[m : n] = [2 : 3]$, one has $3m - 2n = 0$. Formula (15), with the required norm sign, gives $\delta = \alpha^2$ up to sign. Therefore

$$z + y\alpha = \eta\alpha^4 = (\alpha^2 - 2)(3\alpha) = 9 - 6\alpha.$$

Thus

$$(x, y, z) = (\pm 3, -6, 9),$$

which is not primitive.

For $[m : n] = [2 : 7]$, the unscaled triple is

$$(-112, -32, -41).$$

With the required sign, we may take

$$\delta = 112 + 32\alpha + 41\alpha^2.$$

There is a useful internal square identity:

$$\delta = (10 + \alpha + 2\alpha^2)^2.$$

Indeed,

$$(10 + \alpha + 2\alpha^2)^2 = 112 + 32\alpha + 41\alpha^2.$$

Moreover,

$$N(10 + \alpha + 2\alpha^2) = 10^3 + 3 + 9 \cdot 2^3 - 9 \cdot 10 \cdot 1 \cdot 2 = 895.$$

Hence $N(\delta) = 895^2$, so $x = \pm 895$.

A further direct expansion gives

$$\delta^2 = 20416 + 12211\alpha + 10208\alpha^2.$$

Multiplying by $\eta = \alpha^2 - 2$ yields

$$\begin{aligned} \eta\delta^2 &= (3 \cdot 12211 - 2 \cdot 20416) + (3 \cdot 10208 - 2 \cdot 12211)\alpha \\ &= -4199 + 6202\alpha. \end{aligned}$$

Therefore

$$(y, z) = (6202, -4199),$$

and the second unit class contributes exactly

$$(\pm 895, 6202, -4199).$$

8 Completion of the proof

By proposition 4.2, every primitive solution belongs to exactly one of the two unit square classes $e = 0$ or $e = 1$. The first class gives only $(\pm 1, 0, 1)$ after the nonprimitive candidate $(\pm 3, 3, 0)$ is removed. The second gives only $(\pm 895, 6202, -4199)$ after the nonprimitive candidate $(\pm 3, -6, 9)$ is removed. This proves theorem 1.2.

For completeness, the final direct checks are

$$3 \cdot 0^3 + 1^3 = 1 = (\pm 1)^4$$

and

$$3 \cdot 6202^3 + (-4199)^3 = 641641050625 = 895^4.$$

Also

$$\gcd(895, 6202) = 1, \quad \gcd(895, 4199) = 1, \quad \gcd(6202, 4199) = 1,$$

so the large triple is primitive.

A Finite congruence tables

This appendix records the small residue computations used in the proof.

A.1 Squares modulo 16, 27, and 64

The square residues modulo 16 are

$$0, 1, 4, 9.$$

The square residues modulo 27 are

$$0, 1, 4, 7, 9, 10, 13, 16, 19, 22, 25.$$

A square divisible by 16 is congruent to 0 or 16 modulo 64.

A.2 Unit cubes modulo 27

For $u \in (\mathbb{Z}/27\mathbb{Z})^\times$, the possible values of u^3 are

$$1, 8, 10, 17, 19, 26.$$

The expression used in lemma 7.3 has values

t	1	8	10	17	19	26
$18t^2 + 6t + 8 \pmod{27}$	5	20	5	20	5	20.

B Independent exact checks

The following identities can be verified entirely by expansion:

$$\begin{aligned} \text{disc}(X^6 + 60X^3 - 72) &= 2^{18}3^{25}, \\ \text{disc}(-8X^6 - 60X^3 + 9) &= 2^{18}3^{25}, \\ \text{disc}(-F(X, 1)) &= 2^{18}3^{25}, \\ F(-U - V, -2U - V) &= -H(U, V), \\ H(T, 1) - (4T^3 - 3T^2 - 24T + 20)^2 &= 81(T - 1)^3(3T + 5). \end{aligned}$$

The finite-field counts in proposition 5.1 are also obtained by direct enumeration. A companion Python file supplied with the manuscript performs these checks using exact integer and finite-field arithmetic.

C Logical dependency summary

For clarity, the classification rests on the following chain.

1. Primitivity implies pairwise coprimality and excludes the prime 3 from xz .
2. The field $K = \mathbb{Q}(\sqrt[3]{3})$ has integral basis $1, \alpha, \alpha^2$, class number 1, and norm-1 units η^n .
3. The principal ideal $(z + y\alpha)$ is a fourth power, so $z + y\alpha = \eta^e \delta^2$ with $e \in \{0, 1\}$ and $N(\delta) = x^2$.

4. The two values of e produce two explicit conics in the coefficients of δ .
5. Primitive parametrization and local congruences reduce the problem to rational points on C_A, C_B, C_F .
6. The exact genus-two point classification in proposition 5.1 leaves only the four primitive triples stated in theorem 1.2.